

Q#01:-

Sol.

time	s	$\dot{s}$	$\ddot{s}$
$t=0$	$\pi/3$	0	0
$T=4$	$\pi/4$	0	0

$$q(t) = At^5 + Bt^4 + Ct^3 + Dt^2 + Et + F$$

$$\dot{q}(t) = 5At^4 + 4Bt^3 + 3Ct^2 + 2Dt + E$$

$$\ddot{q}(t) = 20At^3 + 12Bt^2 + 6Ct + 2D$$

for $t=0$

$$q(0) = F, \quad \dot{q}(0) = E, \quad \ddot{q}(0) = 2D$$

for $t=4$

$$q(4) = 1024A + 256B + 64C + 16D + 4E + F$$

$$\dot{q}(4) = 1280A + 256B + 48C + 8D + E$$

$$\ddot{q}(4) = 1280A + 192B + 24C + 2D$$

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 2 & 0 & 0 \\ 1024 & 256 & 64 & 16 & 4 & 1 \\ 1280 & 256 & 48 & 8 & 1 & 0 \\ 1280 & 192 & 24 & 2 & 0 & 0 \end{bmatrix} \begin{bmatrix} A \\ B \\ C \\ D \\ E \\ F \end{bmatrix} = \begin{bmatrix} \ddot{v}(0) \\ \dot{v}(0) \\ \ddot{v}(0) \\ \dot{v}(4) \\ \dot{v}(4) \\ \ddot{v}(4) \end{bmatrix}$$

$$\begin{bmatrix} A \\ B \\ C \\ D \\ E \\ F \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \underline{1} \\ 0 & 0 & 0 & 0 & \underline{1} & 0 \\ 0 & 0 & 0 & 2 & 0 & 0 \\ 1024 & 256 & 64 & 16 & 4 & 1 \\ 1280 & 256 & 48 & 8 & 1 & 0 \\ 1280 & 192 & 24 & 2 & 0 & 0 \end{bmatrix}^{-1} \begin{bmatrix} \pi/3 \\ 0 \\ 0 \\ 7\pi/4 \\ 0 \\ 0 \end{bmatrix}$$

$$\text{Inverse} = \begin{bmatrix} -0.0058 & -0.011 & -0.0078 & 0.0058 & -0.011 & 0.0078 \\ 0.058 & 0.125 & 0.093 & -0.058 & 0.109 & -0.0078 \\ -0.15 & -0.31 & -0.375 & 0.156 & -0.25 & 0.125 \\ 0 & 0 & 0.5 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} A \\ B \\ C \\ D \\ E \\ F \end{bmatrix} = \begin{bmatrix} 0.0261 \\ -0.0268 \\ 0.6954 \\ 0 \\ 0 \\ 1.0472 \end{bmatrix}$$

When $t=0$;

$$q = 0.026 \quad v = -0.026 \quad a = 0.6$$

When $t=4$;

$$q = 0 \quad v = 0 \quad a = 1.047$$

$$\therefore \begin{bmatrix} A \\ B \\ C \\ D \\ E \\ F \end{bmatrix} = A^{-1} \begin{bmatrix} \pi/3 \\ 0 \\ 0 \\ 0 \\ \pi/4 \\ 0 \\ 0 \end{bmatrix}$$

Matlab Code:-

$$q1 = \pi/3; \quad s1 = 7*\pi/4; \quad t0 = 0; \quad tf = 4;$$

$$A = \begin{bmatrix} t0^5 & t0^4 & t0^3 & t0^2 & t0 & 1 \\ 5t0^4 & 4t0^3 & 3t0^2 & 2t0 & 1 & 0 \\ 20t0^3 & 12t0^2 & 6t0 & 2 & 0 & 0 \\ tf^5 & tf^4 & tf^3 & tf^2 & tf & 1 \\ 5tf^4 & 4tf^3 & 3tf^2 & 2tf & 1 & 0 \\ 20tf^3 & 12tf^2 & 6tf & 2 & 0 & 0 \end{bmatrix};$$

$$b = [q1; 0; 0; s1; 0; 0];$$

$$A_inv = inv(A);$$

$$val = A_inv * b;$$

$$\text{disp}(A_inv);$$

$$\text{disp}(val);$$

$a_5 = \text{val}(1); \quad a_4 = \text{val}(2); \quad a_3 = \text{val}(3); \quad a_2 = \text{val}(4);$
 $a_1 = \text{val}(5); \quad a_0 = \text{val}(6);$

$\text{fprintf}('q = %.4f \quad v = %.4f \quad a = %.4f', a_5, a_4, a_3);$

$\text{fprintf}('q = %.4f \quad v = %.4f \quad a = %.4f', a_2, a_1, a_0);$

Q#02:-

Sol.

time	s	$\dot{s}$	$\ddot{s}$
$t=0$	$\pi/2$	0	0
$t=6$	$3\pi/2$	0	0

$$q(t) = At^5 + Bt^4 + Ct^3 + Dt^2 + Et + F$$

$$\dot{q}(t) = 5At^4 + 4Bt^3 + 3Ct^2 + 2Dt + E$$

$$\ddot{q}(t) = 20At^3 + 12Bt^2 + 6Ct + 2D$$

for $t=0$

$$q(0) = F, \quad \dot{q}(0) = E, \quad \ddot{q}(0) = 2D$$

for $t=6$

$$q(6) = 7776A + 1296B + 216C + 36D + 6E + F$$

$$\dot{q}(6) = 6480A + 864B + 108C + 12D + E$$

$$\ddot{q}(6) = 4320A + 432B + 36C + 2D$$

$$q_2 = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 2 & 0 & 0 \\ 7776 & 1296 & 216 & 36 & 6 & 1 \\ 6480 & 864 & 108 & 12 & 1 & 0 \\ 4320 & 432 & 36 & 2 & 0 & 0 \end{bmatrix} \begin{bmatrix} A \\ B \\ C \\ D \\ E \\ F \end{bmatrix} = \begin{bmatrix} \pi/2 \\ 0 \\ 0 \\ 3\pi/2 \\ 0 \\ 0 \end{bmatrix}$$

A

B = X

$$B = A^{-1} X$$

$$A^{-1} \Rightarrow \begin{bmatrix} (A_{11}^{-1} - A_{12} A_{22}^{-1} A_{21})^{-1} & -A_{11}^{-1} A_{12} (A_{22} - A_{21} A_{11}^{-1} A_{12})^{-1} \\ -A_{22}^{-1} A_{21} (A_{11} - A_{12} A_{22}^{-1} A_{21})^{-1} & (A_{22} - A_{21} A_{11}^{-1} A_{12})^{-1} \end{bmatrix}$$

$$A_{11} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad A_{12} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 2 & 0 & 0 \end{bmatrix} \quad A_{21} = \begin{bmatrix} 7776 & 1296 & 216 \\ 6480 & 864 & 108 \\ 4320 & 432 & 36 \end{bmatrix}$$

$$A_{22} = \begin{bmatrix} 36 & 6 & 1 \\ 12 & 1 & 0 \\ 2 & 0 & 0 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} -0.0008 & -0.0023 & -0.0023 & 0.0008 & -0.0023 & 0.0008 \\ 0.0116 & 0.0370 & 0.417 & -0.0116 & 0.0324 & -0.0216 \\ -0.0463 & -0.1667 & -0.25 & 0.0463 & -0.111 & 0.0833 \\ 0 & 0 & 0.5 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$B = A^{-1} X$$

$$B = A^{-1} \begin{bmatrix} \bar{T}_2 \\ 0 \\ 0 \\ 3\bar{T}_2 \\ 0 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} A \\ B \\ C \\ D \\ E \\ F \end{bmatrix} = A^{-1} \begin{bmatrix} \bar{T}_2 \\ 0 \\ 0 \\ 3\bar{T}_2 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} A \\ B \\ C \\ D \\ E \\ F \end{bmatrix} = \begin{bmatrix} 0.0024 \\ -0.0364 \\ 0.1454 \\ 0 \\ 0 \\ 1.5708 \end{bmatrix}$$

When $\bar{T} = 0$

$$q = 0.0024 \quad v = -0.0364 \quad a = 0.145$$

When $\bar{T} = 6$

$$q = 0 \quad v = 0 \quad a = 1.5708$$

$$q(4) = 45207 + 45207 - \dots$$

Q#03:-

Sol.

time	s	$\dot{s}$	$\ddot{s}$
$t = 3$	0	2×10^{-9}	0
$t = 8$	$\pi/2$	5×10^{-9}	0

$$q(t) = At^5 + Bt^4 + Ct^3 + Dt^2 + Et + F$$

$$\dot{q}(t) = 5At^4 + 4Bt^3 + 3Ct^2 + 2Dt + E$$

$$\ddot{q}(t) = 20At^3 + 12Bt^2 + 6Ct + 2D$$

for $t = 3$

$$q(3) = 243A + 81B + 27C + 9D + 3E + F$$

$$\dot{q}(3) = 405A + 108B + 27C + 6D + E$$

$$\ddot{q}(3) = 540A + 108B + 18C + 2D$$

for $t = 8$

$$q(8) = 32768A + 4096B + 512C + 64D + 8E + F$$

$$\dot{q}(8) = 20480A + 2048B + 192C + 16D + E$$

$$\ddot{q}(8) = 10240A + 768B + 48C + 2D$$

$$\begin{bmatrix} A \\ B \\ C \\ D \\ E \\ F \end{bmatrix} = \begin{bmatrix} 243 & 81 & 27 & 9 & 3 & 1 \\ 405 & 108 & 27 & 6 & 1 & 0 \\ 540 & 108 & 18 & 2 & 0 & 0 \\ 32768 & 4096 & 512 & 64 & 8 & 1 \\ 20480 & 2048 & 192 & 16 & 1 & 0 \\ 10240 & 768 & 48 & 2 & 0 & 0 \end{bmatrix}^{-1} \begin{bmatrix} 0 \\ 2 \times 10^{-9} \\ 0 \\ \frac{\pi}{2} \\ 5 \times 10^{-9} \\ 0 \end{bmatrix}$$

$X = A^{-1} B$

$$A^{-1} = \begin{bmatrix} -0.0019 & -0.0048 & -0.0040 & 0.0019 & -0.0048 & 0.0040 \\ 0.0528 & 0.1360 & 0.1200 & -0.0528 & 0.1280 & -0.100 \\ 0.5408 & -1.4400 & -1.3800 & 0.5408 & -1.2640 & 0.9400 \\ 2.5344 & 6.9120 & 7.5200 & -2.5344 & 5.760 & -4.1400 \\ -5.5296 & -14.336 & -19.200 & 5.529 & -12.312 & 8.6400 \\ 5.5706 & 9.830 & 18.432 & -4.570 & 10.022 & -6.9120 \end{bmatrix}$$

$$\begin{bmatrix} A \\ B \\ C \\ D \\ E \\ F \end{bmatrix} = A^{-1} \begin{bmatrix} 0 \\ 2 \times 10^{-9} \\ 0 \\ \frac{\pi}{2} \\ 5 \times 10^{-9} \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} A \\ B \\ C \\ D \\ E \\ F \end{bmatrix} = \begin{bmatrix} 0.0030 \\ -0.0829 \\ 0.8495 \\ -3.9810 \\ 8.6859 \\ -7.1794 \end{bmatrix}$$

When $T=3$

$$q_1 = 0.0030 \quad v = -0.0829 \quad a = 0.8495$$

When $T=8$

$$q_1 = -3.9810 \quad v = 8.6859 \quad a = -7.1794$$

Matlab Code:-

$$q_1 = 0;$$

$$s_1 = p_1/2;$$

$$t_0 = 3;$$

$$t_f = 8;$$

$$q_2 = 2 \times 10^{-9};$$

$$s_2 = 5 \times 10^{-9};$$

$$A = \begin{bmatrix} t_0^5 & t_0^4 & t_0^3 & t_0^2 & t_0 & 1 \\ 5 \cdot t_0^4 & 4 \cdot t_0^3 & 3 \cdot t_0^2 & 2 \cdot t_0 & 1 & 0 \\ 20 \cdot t_0^3 & 12 \cdot t_0^2 & 6 \cdot t_0 & 2 & 0 & 0 \\ t_f^5 & t_f^4 & t_f^3 & t_f^2 & t_f & 1 \\ 5 \cdot t_f^4 & 4 \cdot t_f^3 & 3 \cdot t_f^2 & 2 \cdot t_f & 1 & 0 \\ 20 \cdot t_f^3 & 12 \cdot t_f^2 & 6 \cdot t_f & 2 & 0 & 0 \end{bmatrix};$$

$$b = [q_1; q_2; 0; s_1; s_2; 0];$$

```
A_inv = inv(A);  
disp(A_inv);  
val = A_inv * b;  
disp(val);
```

```
a5 = val(1); a4 = val(2); a3 = val(3); a2 = val(4);
```

```
a1 = val(5); a0 = val(6);
```

```
fprintf('q = %.4f   v = %.4f   a = %.4f', a5, a4, a3);
```

```
fprintf('q = %.4f   v = %.4f   a = %.4f', a2, a1, a0);
```

Section - C:-

* Printed Copy Attached

Section - B:-

Q# 01:-

Given Data:-

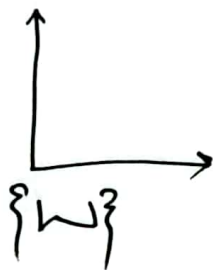
$$x = 4.7343, y = -3.8242, \theta = 160$$

Transformation Matrix:-

$$T_F = \begin{bmatrix} \cos(160) & -\sin(160) & 4.7343 \\ \sin(160) & \cos(160) & -3.8242 \\ 0 & 0 & 1 \end{bmatrix}$$

→ Final Transformation Matrix

Transformation Matrix for adjacent frames :



$$T_0 = \begin{pmatrix} \cos 0 & -\sin 0 & 4 \\ \sin 0 & \cos 0 & 2 \\ 0 & 0 & 1 \end{pmatrix}$$

→ Translation Matrix

Translation is Through (4, 2).

blw frame W and the next frame.

$$T_1 = \begin{pmatrix} \cos \theta_1 & -\sin \theta_1 & 3 \\ \sin \theta_1 & \cos \theta_1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Between $\{2\}$ & $\{1\}$

$$T_2 = \begin{pmatrix} \cos \theta_2 & -\sin \theta_2 & d_2 \\ \sin \theta_2 & \cos \theta_2 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Between $\{3\}$ & $\{4\}$

$$T_3 = \begin{pmatrix} \cos(40) & -\sin(40) & 4 \\ \sin(40) & \cos(40) & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Between $\{4\}$ & $\{5\}$

$$T_4 = \begin{pmatrix} \cos(30) & -\sin(30) & 0 \\ \sin(30) & \cos(30) & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Last Link

$$T_0 \times T_1 \times T_2 \times T_3 \times T_4 = T_F$$

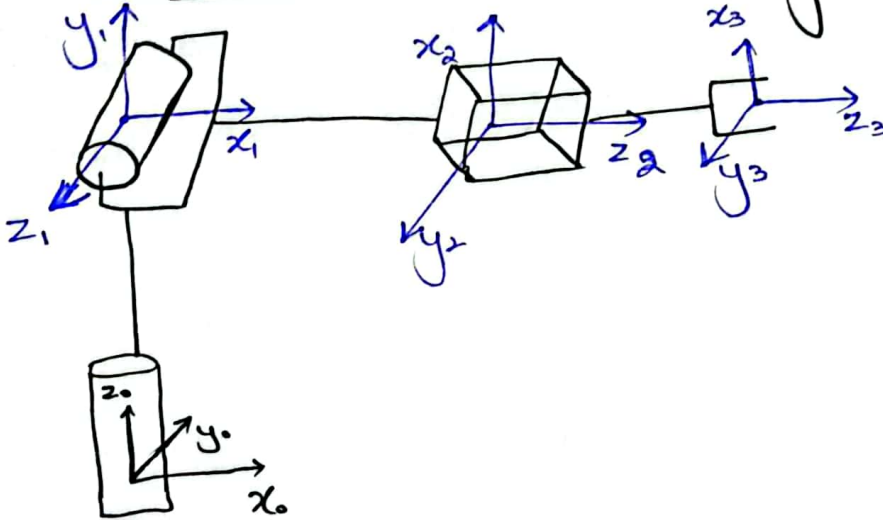
$$T_4^0 = T_F$$

Q#02:-

Given Data:-

$$x=3, y=4, \theta=0$$

DH-Table ? Framing ?



DH Table:-

θ	d	a	α
θ_1	a_1	0	90°
$\theta_2 + 90$	0	a_2	90°
0	d_3	a_3	0

$$T = \begin{pmatrix} \cos \theta & -\sin \theta \cos \alpha & \sin \theta \sin \alpha & a \cos \theta \\ \sin \theta & \cos \theta \cos \alpha & -\cos \theta \sin \alpha & a \sin \theta \\ 0 & \sin \alpha & \cos \alpha & d \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$T_0 = \begin{pmatrix} \cos \theta_1 & 0 & \sin \theta_1 & 0 \\ \sin \theta_1 & 0 & -\cos \theta_1 & 0 \\ 0 & 1 & 0 & 5 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$T_1 = \begin{pmatrix} \cos(\theta_2 + 90) & 0 & \sin(\theta_2 + 90) & 5 \cos(\theta_2 + 90) \\ \sin(\theta_2 + 90) & 0 & -\cos(\theta_2 + 90) & 5 \sin(\theta_2 + 90) \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$T_2 = \begin{pmatrix} 1 & 0 & 0 & 4 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$T_1 \times T_2 \times T_3 = T_F$$

By Multiplying T_1 & T_2

$$\begin{pmatrix} \cos \theta_1 \cos(\theta_2 + 90) & \sin \theta_1 \cos(\theta_2 + 90) & \cos \theta_1 \sin(\theta_2 + 90) & 5 \cos \theta_1 \cos(\theta_2 + 90) \\ \sin \theta_1 \cos(\theta_2 + 90) & -\cos \theta_1 \cos(\theta_2 + 90) & \sin \theta_1 \sin(\theta_2 + 90) & 5 \sin \theta_1 \cos(\theta_2 + 90) \\ \sin(\theta_2 + 90) & 0 & -\cos(\theta_2 + 90) & 5 + 5 \sin(\theta_2 + 90) \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$T_1 \times T_2 = T'$$

$$T^* \times T_3 = \begin{bmatrix} \cos \theta_1 \cos(\theta_2 + 90) & -\cos \theta_1 \sin(\theta_2 + 90) & 0 & 5 \cos \theta_1 \cos \theta_2 + 4 \cos \theta_1 \\ \sin \theta_1 \cos(\theta_2 + 90) & -\sin \theta_1 \sin(\theta_2 + 90) & 0 & 5 \sin \theta_1 \cos \theta_2 + 4 \sin \theta_1 \\ 0 & 0 & 1 & 5 + d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$5 \cos \theta_1 \cos(\theta_2 + 90) + 4 \cos \theta_1 = 3 \quad \text{--- (1)}$$

$$5 \sin \theta_1 \cos(\theta_2 + 90) + 4 \sin \theta_1 = 4 \quad \text{--- (2)}$$

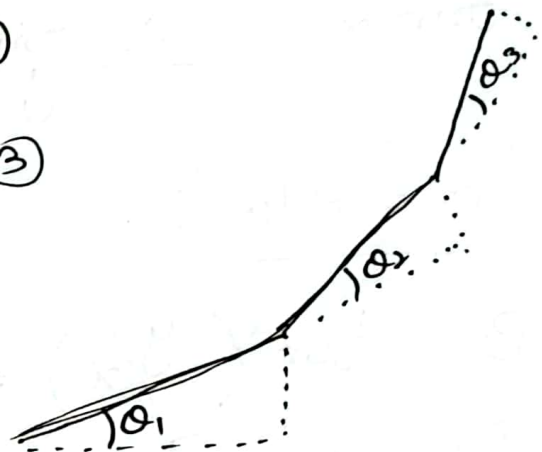
$$5 + d_3 = 0 \quad \text{--- (3)}$$

$$\boxed{d_3 = -5} \quad \text{from (3)}$$

Q#03:-

Given Data:-

$$x=5, \quad y=7, \quad \theta=45^\circ$$



$$x = a_1 \cos \theta_1 + a_2 \cos(\theta_1 + \theta_2) + a_3 (\cos(\theta_1 + \theta_2 + \theta_3))$$

$$y = a_1 \sin \theta_1 + a_2 \sin(\theta_1 + \theta_2) + a_3 \sin(\theta_1 + \theta_2 + \theta_3)$$

$$\theta = \theta_1 + \theta_2 + \theta_3 \Rightarrow \theta_3 = 45^\circ - (\theta_1 + \theta_2)$$

$$d = \sqrt{x^2 + y^2}$$

$$= \sqrt{74} = 8.6$$

$$\cos \theta_2 = \frac{a_1^2 + a_2^2 - d^2}{2a_1 a_2}$$

$$= \frac{(5)^2 + (4)^2 - (8.6)^2}{2(5)(4)} = -0.825$$

$$\theta_2 = \cos^{-1}(-0.825) = \boxed{145.53^\circ}$$

$$\sin \alpha = \frac{a_2 \sin \theta_2}{d} \Rightarrow \sin^{-1}\left(\frac{4 \sin(145.53)}{8.6}\right)$$

$$\boxed{\alpha = 24.62}$$

$$\beta = \tan^{-1}(y/x) = \tan^{-1}\left(\frac{7}{5}\right)$$

$$\boxed{\beta = 54.46}$$

$$\theta_1 = \beta - \alpha$$

$$\theta_1 = 54.46 - 24.62$$

$$\boxed{\theta_1 = 29.84}$$

$$\theta_3 = 45^\circ - 29.84^\circ - 145.53^\circ$$

$$\boxed{\theta_3 = -130.37^\circ}$$

Q# 04:-

DH Table:-

θ	d	a	α
θ_1	d_1	0	0
0	d_2	0	-90
0	d_3	0	0

$$T_0 = \begin{bmatrix} \cos \theta_1 & -\sin \theta_1 & 0 & 0 \\ \sin \theta_1 & \cos \theta_1 & 0 & 0 \\ 0 & 0 & 1 & d_1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T_1 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & -d_2 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & \underline{d} \end{bmatrix}$$

$$T_2 = \begin{bmatrix} 1 & 0 & 0 & d_3 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T_F = T_0 \times T_1 \times T_2$$

$$T_F = \begin{pmatrix} \cos \theta_1 & 0 & \sin \theta_1 & -d_2 \sin \theta_1 + d_3 \cos \theta_1 \\ \sin \theta_1 & 0 & -\cos \theta_1 & d_2 \cos \theta_1 + d_3 \sin \theta_1 \\ 0 & 1 & 0 & d_1 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$x = -d_2 \sin \theta_1 + d_3 \cos \theta_1$$

$$y = d_2 \cos \theta_1 + d_3 \sin \theta_1$$

$$z = d_1$$

SEC-A Q# 01: (OUTPUT)

Command Window

```
>> ass1
```

Inverse of A:

Command Window

```
>> ass1
```

Inverse of A:

-0.0059	-0.0117	-0.0078	0.0059	-0.0117	0.0078
0.0586	0.1250	0.0938	-0.0586	0.1094	-0.0625
-0.1563	-0.3750	-0.3750	0.1563	-0.2500	0.1250
0	0	0.5000	0	0	0
0	1.0000	0	0	0	0
1.0000	0	0	0	0	0

Quintic polynomial coefficients:

0.0261
-0.2608
0.6954
0
0
1.0472

WHEN T=0

q= 0.0261

v= -0.2608

a= 0.6954

WHEN T=4

f_A q= 0.0000

v= 0.0000

a= 1.0472>>

SEC-A Q# 03: (OUTPUT)

Command Window

```
>> ass1
```

```
s2 =
```

5.0000e-09

Inverse of A:

-0.0019	-0.0048	-0.0040	0.0019	-0.0048	0.0040
0.0528	0.1360	0.1200	-0.0528	0.1260	-0.1000
-0.5408	-1.4400	-1.3800	0.5408	-1.2640	0.9400
2.5344	6.9120	7.5200	-2.5344	5.7600	-4.1400
-5.5296	-14.3360	-19.2000	5.5296	-12.3120	8.6400
5.5706	9.8304	16.4320	-4.5706	10.0224	-6.9120

Quintic polynomial coefficients:

0.0030
-0.0829
0.8495
-3.9810
8.6859
-7.1794

WHEN T=3

q= 0.0030

v= -0.0829

a= 0.8495

WHEN T=8

f_A q= -3.9810

v= 8.6859

a= -7.1794>> |

```

#include <ESP8266WiFi.h>

const char* ssid = "your_ssid";
const char* password = "your_password";
int array[] = {D1, D2, D3, D4, D5};

int var = 0;
WiFiServer server(80);

void setup()
{
    Serial.begin(115200);

    for (int i = 0; i < 5; i++)
    {
        pinMode(array[i], OUTPUT);
        digitalWrite(array[i], LOW);
    }

    WiFi.begin(ssid, password);
    while (WiFi.status() != WL_CONNECTED)
    {
        delay(100);
        Serial.print(".");
    }
    Serial.println();
    Serial.println(WiFi.localIP());
    server.begin();
}

void loop()
{
    WiFiClient client = server.available();
    if (!client)
    {
        return;
    }

    while (!client.available())
    {
        delay(10);
    }

    String str = client.readStringUntil('\r');
    Serial.println(str);
}

```

```

client.flush();
if (str.indexOf("/pattern1") != -1)
{
    var = 1;
}
else if (str.indexOf("/pattern2") != -1)
{
    var = 2;
}
else if (str.indexOf("/pattern3") != -1)
{
    var = 3;
}

client.println("HTTP/1.1 200 OK");
client.println("Content-Type: text/html");
client.println();
client.println("<!DOCTYPE HTML>");
client.println("<html>");
client.println("<h1>Wedding Lights Control</h1>");
client.println("<button onclick=\"location.href=' /pattern1'\">Pattern
1</button>");
client.println("<button onclick=\"location.href=' /pattern2'\">Pattern
2</button>");
client.println("<button onclick=\"location.href=' /pattern3'\">Pattern
3</button>");
client.println("</html>");
client.stop();

switch (var)
{
    case 1:
        pattern1();
        break;
    case 2:
        pattern2();
        break;
    case 3:
        pattern3();
        break;
}
}

void pattern1()

```

```
for (int i = 0; i < 5; i++)
{
    digitalWrite(array[i], HIGH);
    delay(10);
    digitalWrite(array[i], LOW);
}

void pattern2()
{
    for (int i = 4; i >= 0; i--) {
        digitalWrite(array[i], HIGH);
        delay(10);
        digitalWrite(array[i], LOW);
    }
}

void pattern3()
{
    for (int i = 0; i < 5; i++)
    {
        digitalWrite(array[i], HIGH);
    }
    delay(50);
    for (int i = 0; i < 5; i++)
    {
        digitalWrite(array[i], LOW);
    }
    delay(50);
}
```